

Javad Omidi

Postdoctoral Associate, Cornell University, Ithaca, NY, US

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U.S. Permanent Resident (Green Card Holder)

PROFESSIONAL SUMMARY

Computational and experimental engineering researcher with 15+ years of experience in numerical simulation, fluid dynamics, multiphysics modeling, optimization, and data-driven research. Extensive expertise in CFD, CFD-DEM, CFD-TFM, multiphase and turbulent flows, granular materials, soft matter, EHD, micro-plasma actuators, aerodynamic flow control, and wind turbine and renewable-energy applications. Experienced in numerical algorithm development and improvement, parametric studies and optimization, HPC, experimental validation, optical and MRI-based imaging, and quantitative flow characterization. Complementary expertise in large-scale data analytics, machine learning, deep learning, and predictive modeling using Python/R.

ACADEMICS

MAR 2026	Postdoctoral Associate
MAR 2025	Cornell University, Ithaca, NY, US Computational & Data Analytics Research
FEB 2025	PhD, “Chemical Engineering”
JAN 2022	Columbia University, New York City, NY, US CGPA: 4.0/4.0 Supervisor: Prof. Christopher Boyce PhD Thesis: <i>In Silico Simulation Studies of Structured Gas-Fluidized Granular Flows</i>
DEC 2023	MPhil, “Chemical Engineering”
JAN 2022	Columbia University, New York City, NY, US CGPA: 4.0/4.0 Supervisor: Prof. Chris Boyce
JAN 2015	MS, “Mechanical Engineering”
SEP 2012	Sharif University of Technology, Tehran, Iran CGPA: 3.8/4.0 MS Thesis: <i>Numerical Simulation and Optimization of the DBD Plasma Actuator in Boundary Layer Control</i> Supervisor: Prof. Karim Mazaheri
SEP 2012	BS, “Mechanical Engineering”
SEP 2007	Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran CGPA: 3.2/4.0 BS Thesis: <i>Computational Fluid Dynamic Study on Empirical Correlation of the Axial-Flow Compressor Cascades</i> Supervisor: Prof. Abolghasem Mesgarpour Tousi
SEP 2007	High School & Pre-University, “Mathematics and Physics”
SEP 2003	Farhangian High School, Arak, Iran CGPA: 4.0/4.0

RESEARCH PROJECTS

MAR 2027	High-Dimensional Data Analysis and Predictive Modeling
MAR 2026	Cornell University, Ithaca, NY, US - Analyzed large-scale, high-dimensional datasets - Developed automated data-processing pipelines - Built ML classification and feature-ranking models - Integrated heterogeneous data sources - Performed statistical and comparative analyses - Developed reproducible Python/R workflows

JAN 2026
JAN 2022

Large-Scale Data Analytics and Machine Learning

Columbia University, New York City, NY, US

- Integrated large-scale, high-dimensional datasets
- Developed ML classification and feature-selection pipelines
- Applied autoencoders for latent representation learning
- Constructed and analyzed complex interaction networks
- Performed multivariate statistical and predictive modeling
- Built reproducible Python/R data-analysis workflows

SEP 2025
SEP 2024

Advanced CFD-DEM Simulations of Structured Bubbling in Fluidized Beds under Oscillated Gas Flow & Horizontal Vibration

Supervisor: Prof. Chris Boyce, Columbia University, New York City, NY, US

- Horizontal vibration enhances particle mixing efficiency
- Structured bubbling preserved at frequency ratio 3–5
- Optimal vibration strength near unity for stable bubbling
- Excessive vibration disrupts structured bubble lattices
- Oscillated gas flow induces triangular bubble lattice
- Structured bubbling persists under particle size heterogeneity
- Larger bubbles improve drag and mixing performance
- Gas oscillation prevents de-fluidization of dense particles
- Industrial potential for vibrated structured bubbling reactors

JUN 2025
NOV 2024

Structured Bubbling Control in Fluidized Beds via Oscillating Gas Injection and Phase-Offset Modeling

Supervisor: Prof. Chris Boyce, Columbia University, New York City, NY, US

- Structured bubbling via oscillated gas flow
- Phase-offset slicing enables diverse bubble lattices
- Enhanced horizontal particle mixing with structured bubbling
- Local particle solidification drives bubble structuring
- Bubble size reduced by distributor slicing
- Optimal mixing at $\pi/2$ phase offset
- Wide variation of bubble structures in simulations
- High particle pressure confirms structural stability
- Faster mixing than conventional bubbling methods
- Framework for predictive machine learning bubble models

JAN 2025
JAN 2024

Anomalous Descent of Intruders in Vibrated Gas-Fluidized Granular Materials

Supervisor: Prof. Chris Boyce, Columbia University, New York City, NY, US

- Oscillatory descent velocity of dense intruders
- Void fraction fluctuations control intruder drag force
- CFD-DEM simulations validate experimental descent dynamics
- Intruder size effects on descent velocity anomalies
- Drafting-kissing-tumbling analog in granular systems
- Bottom intruder slowdown from particle compaction
- Stable side-by-side intruder descent observed
- Locally varying rheology in vibrated gas-fluidized grains

AUG 2024
JAN 2022

Magnetic Resonance Imaging for 3D Characterization of Bubble Dynamics in Dense Liquid–Solid Suspensions

Supervisor: Prof. Chris Boyce, Columbia University, New York City, NY, US

- 3D MRI of bubble streams in dense suspensions
- High-resolution multi-band echo planar imaging (MB-EPI)
- Non-invasive visualization of bubble rise and coalescence
- Bubble dynamics depend on particle volume fraction
- Idle time controls frequency of bubble coalescence
- Coalescence increases bubble size but slows rise velocity
- Non-sphericity significantly impacts bubble velocity predictions
- Effective viscosity overestimated in high particle fractions

SEP 2023
AUG 2022

Faraday Wave Dynamics and Particle Mixing in Gas-Fluidized Beds under Combined Vertical and Horizontal Vibrations

Supervisor: Prof. Chris Boyce, Columbia University, New York City, NY, US

- Two-fluid modeling of Faraday waves in fluidized beds
- Combined vertical–horizontal vibration enhances particle mixing
- Horizontal vibration increases horizontal mixing efficiency
- Vertical frequency decreases wavelength and wave height
- Gas flow increases wavelength but reduces wave height
- Phase offset has minor effect on wave dynamics
- Horizontal vibration alone cannot form surface waves
- Regime maps and dimensionless correlations for wave prediction
- Optimal mixing at low vertical and high horizontal frequency

JAN 2022
MAY 2021

Optimization of Plasma Actuator Placement on Curved Aerodynamic Surfaces Using Differential Evolution

Supervisor: Prof. Karim Mazaheri, Sharif University of Technology, Tehran, Iran

- DBD plasma actuator optimization using Differential Evolution
- Improved electrostatic model for actuator–flow interaction
- Optimal placement between 2–4% chord from leading edge
- Lift coefficient increased by over 30% at stall
- Stall delay up to 26% with optimal location
- Mathematical model linking Reynolds number and angle of attack
- Flow separation control validated with electrohydrodynamic solver
- Regime maps for actuator placement under varied conditions

JAN 2022
APR 2021

Swimming Cylinder Wake Control with Plasma Actuator

Supervisor: Prof. Karim Mazaheri, Sharif University of Technology, Tehran, Iran

- DBD plasma actuator wake control at $Re = 20,000$
- Enhanced electrostatic model for actuator–turbulent flow interaction
- Actuator placement at 90° improves wake stability
- Drag coefficient reduced by up to 90%
- Lift coefficient fluctuations reduced by 40–55%
- Superior performance of 90° vs 45° placement
- Tandem cylinders: downstream flow improved by 20–26%
- Actuator voltage tuning enhances downstream flow quality

MAR 2021
APR 2020

Plasma Actuator–Driven Aerodynamic Enhancement of Large-Scale Wind Turbines

Supervisor: Prof. Karim Mazaheri, Sharif University of Technology, Tehran, Iran

- DBD plasma actuators for vehicle drag reduction
- Electrohydrodynamic modeling of plasma–flow interaction
- Actuator placement strategy for optimal aerodynamic performance
- Drag coefficient reduction up to 9% in simulations
- Improved flow separation control around vehicle surfaces
- Numerical validation with Navier–Stokes solvers
- Energy-efficient plasma actuation for ground vehicles
- Framework for real-time actuator optimization

MAR 2020
JAN 2018

Parametric Studies of Micro-Plasma Actuators for Aerodynamic Performance of Wind Turbine Blades

Supervisor: Prof. Karim Mazaheri, Sharif University of Technology, Tehran, Iran

- Geometrical and operational dual-parameter study
- Plasma actuator optimization for wind turbine aerodynamics
- Improved electrostatic model validated for flow control design
- Geometrical parameters: dielectric, electrode thickness, electrode length
- Operational parameters: applied voltage, frequency, waveform selection
- Dielectric permittivity and electrode length with saturation limits
- Higher voltage improves lift up to 130%
- Frequency increase beneficial until optimal threshold
- Rectangular waveform yields superior lift performance
- Integrated geometric and operational parametric analysis
- Engineering formulas for optimum actuator configuration

APR 2020
JAN 2018

Differential Evolution Optimization of Micro Plasma Actuators for Aerodynamic Performance Enhancement

Supervisor: Prof. Karim Mazaheri, Sharif University of Technology, Tehran, Iran

- Plasma actuator optimization for wind turbine blades
- Improved phenomenological electrostatic model
- Coupled Maxwell–Navier–Stokes numerical framework
- Differential Evolution global optimization algorithm
- Geometrical parameters: electrode length, thickness, dielectric permittivity
- Operational parameters: applied voltage, frequency, waveform
- Lift-to-drag ratio optimization under stall conditions
- Optimum dielectric and electrode thickness thresholds identified
- Engineering design formula for actuator configuration
- Validated with airfoil experiments and simulations

JAN 2018
SEP 2013

Novel Algorithmic Advancements in Numerical Modeling of DBD Plasma Actuators

Supervisor: Prof. Karim Mazaheri, Sharif University of Technology, Tehran, Iran

- Modified phenomenological model for DBD plasma actuators
- New boundary conditions for electric potential and charge density
- Semi-empirical relations for Debye length and plasma extent
- Accurate wall-jet and boundary layer simulation without calibration
- Aerodynamic performance improvement of wind turbine airfoils
- Flow separation control at high angle of attack
- Optimum actuator placement near separation onset
- Tandem actuator configuration analysis and performance gain
- Wide-range validation with experimental data and simulations
- Numerical framework linking electrostatics and Navier–Stokes solvers

RESEARCH, TEACHING, & ADVISING EXPERIENCES

MAR 2027
MAR 2026

Postdoctoral Associate

Cornell University, Ithaca, NY, US

Conducting data-driven computational research focused on large-scale, high-dimensional datasets. Developing Python/R workflows for data processing, integration, statistical analysis, and visualization, with emphasis on machine learning, feature selection, classification, predictive modeling, network analysis, and comparative data analysis across heterogeneous datasets.

AUG 2026
JUL 2026

STEM Research Mentor

Herricks High School, New Hyde Park, NY, US

Mentored a high school student on an independent data-driven research project involving large-scale dataset analysis, statistical comparison, and feature identification. Provided training in Python-based data analysis, research methodology, statistical methods, literature review, and scientific writing.

NOW
OCT 2016

Scientific Peer-Reviewed Journals

- AIAA Journal (American Institute of Aeronautics and Astronautics)
- Journal of Applied Physics (American Institute of Physics)
- Physics of Fluids (American Institute of Physics)
- Chemical Engineering Science (Elsevier)
- Chemical Engineering Journal (Elsevier)
- Powder Technology (Elsevier)
- IEEE Access (Institute of Electrical and Electronics Engineers)
- Heliyon (Cell Press)
- Scientific Reports (Springer Nature)
- Energy Engineering (Tech Science Press)
- Wind Engineering (Sage)

DEC 2025
JAN 2022

Research Assistant (RA)

PI: Prof. Chris Boyce, Department of Chemical Engineering, Columbia University, New York City, NY, US

Conducted experimental and computational research in granular flow, soft matter, and multi-phase systems. Developed CFD–DEM and CFD–TFM simulations of gas–solid fluidized beds and particle–fluid dynamics using HPC, complemented by optical imaging, image processing, and 3D MRI for experimental characterization of complex multiphase flows.

DEC 2025 JAN 2022	Independent Researcher Department of Chemical Engineering, Columbia University, New York City, NY, US Conducted independent research in large-scale data analytics, machine learning, and high-dimensional data integration. Developed Python/R computational pipelines for feature selection, classification, deep-learning-based representation, network analysis, and multivariate predictive modeling across heterogeneous datasets.
MAY 2023 JAN 2022	Teaching Assistant (TA) Department of Chemical Engineering, Columbia University, New York City, NY, US Graduate Course, Transport in Fluid Mixtures, Prof. Chris Durning — Spring 2023 Graduate Course, Math Methods in Chemical Engineering, Prof. Venkat Venkatasubramanian — Fall 2022 Graduate Course, Advanced Chemical Kinetics, Dr. Robert Bozic — Spring 2022 Delivered computational-modeling modules and supervised project-based learning for graduate cohorts in fluid transport, kinetics, and mathematical modeling, enhancing course engagement metrics
JAN 2023	Workshop Organizer Team 18th Northeast Complex Fluids and Soft Matter Workshop Department of Chemical Engineering, Columbia University, NY, US Coordinated logistics, scientific programming, and panel sessions for 200+ attendees, fostering cross-institutional collaboration in soft-matter and granular-materials research
DEC 2021 FEB 2015	Research Assistant (RA) PI: Prof. Karim Mazaheri, Sharif University of Technology, Tehran, Iran Conducted computational research in CFD, turbulent flow, EHD, and renewable energy, focusing on micro-plasma actuator modeling and aerodynamic flow control. Developed and improved numerical algorithms for coupled plasma–fluid simulations and applied optimization methods to enhance actuator performance, flow separation control, and wind turbine aerodynamics.
JAN 2015 FEB 2014	Teaching Assistant (TA) Sharif University of Technology, Tehran, Iran Undergraduate Course, Aerodynamics I — Fall 2014 Graduate Course, CFD I — Fall 2014 Graduate Course, Wind Turbine Aerodynamics — Fall 2013 Graduate Course, Aerodynamics II — Fall 2013 Graduate Course, CFD I — Fall 2013 Undergraduate Course, Heat Transfer — Spring 2013
JAN 2017 DEC 2015	Thesis Advisor Sharif University of Technology, Tehran, Iran MS Thesis Title: Numerical Simulation of DBD Plasma Actuator and Optimization with Differential Evolution (DE) Algorithm for Separation Control, By: Sajjad Jafari, Supervisor: Prof. Karim Mazaheri MS Thesis Title: Aerodynamic Performance Improvement of Megawatt Wind Turbine Rotor Using Plasma Actuator, By: Mohammadreza Movahhedi, Supervisor: Dr. Abbas Ebrahimi Designed the research scope and methodology and directly advised both MS theses on CFD simulation, DBD plasma actuator modeling, aerodynamic flow control, optimization, and numerical analysis.
OCT 2014	Training Workshop on C++ Programming & Application in CFD Sharif University of Technology, Tehran, Iran
SEP 2015 SEP 2014	Educational/Emotional Counseling and Academic Planning for University Entrance Exam Applicants Arak Library, Arak, Iran

JOURNAL PUBLICATIONS

- (2025)
Published **Structured Bubbling in Fluidized Beds Subjected to Oscillated Gas Velocity and Horizontal Vibration: CFD-DEM Simulations**
Chemical Engineering Science
Javad Omid, Christopher M. Boyce
- (2025)
Published **Structured bubbling flow in fluidized beds with oscillating gas injection which alternates with horizontal position**
AIChE Journal
Javad Omid, Christopher M. Boyce
- (2025)
Published **Effects of distributions of particle size and density on structured bubbling in fluidized beds: CFD-DEM simulations**
Chemical Engineering Science
Oscar J Punch, Javad Omid, Christopher M Boyce
- (2025)
Published **Anomalous descent of intruders in vibrated gas-fluidized granular materials**
Physical Review E
Oscar J. Punch, Qiang Guo, Maty Gueye, Javad Omid, Michael W. Jordan, Jagan Mohan Sanghishetty, Christopher M. Boyce
- (2024)
Published **Magnetic resonance imaging of a stream of bubbles injected into liquid suspensions**
Chemical Engineering Journal
Alireza Bordbar, Wasif Zia, Janine Birnbaum, Javad Omid, Ray F. Lee, Einat Lev, Christopher M. Boyce
- (2024)
Published **Micro Plasma Actuator Location Effect on Complex Fluidic Behavior: An Optimization Study**
arXiv preprint arXiv:2407.06021
Javad Omid
- (2024)
Published **Swimming Cylinder Wake Control with Plasma Actuator**
arXiv preprint arXiv:2407.04876
Javad Omid
- (2024)
Published **Faraday waves in gas-fluidized beds subject to combined vertical and horizontal vibration**
Powder Technology
Javad Omid, Oscar J. Punch, Qiang Guo, Christopher M. Boyce
- (2024)
Published **Advances and opportunities in wind energy harvesting using plasma actuators: a review**
Clean Energy
Javad Omid
- (2023)
Published **Sensitivity Analysis and Parametric Optimization of Micro-Plasma Actuators: A Mini Review**
arXiv preprint arXiv:2308.08437
Javad Omid
- (2023)
Published **Micro-plasma actuator mechanisms in interaction with fluid flow for wind energy applications: operational parameters**
Engineering with Computers
Javad Omid, Karim Mazaheri
- (2022)
Published **DBD plasma actuator effect on mid-blade aerodynamics enhancement of a NREL 5-MW wind turbine**
IEEE Xplore
Javad Omid

- (2021)
Published **Aerodynamic enhancement and improving the performance of a six-megawatt DOWEC wind turbine by micro-plasma actuator**
International Journal of Mechanical Sciences
Javad Omid, Karim Mazaheri
- (2020)
Published **Differential evolution algorithm for performance optimization of the micro plasma actuator as a microelectromechanical system**
Nature Scientific Reports
Javad Omid, Karim Mazaheri
- (2020)
Published **Micro-plasma actuator mechanisms in interaction with fluid flow for wind energy applications: physical parameters**
Physics of Fluids
Javad Omid, Karim Mazaheri
- (2020)
Published **Wind turbine generated power enhancement by plasma actuator**
arXiv preprint arXiv:2004.07046
Javad Omid, Karim Mazaheri
- (2020)
Published **EHD Simulation of Plasma Actuator Effect on Fluid Flow with Boundary Condition Modification in Suzen-Huang Model**
Journal of Iranian Mechanical Engineering Transactions of the ISME
Javad Omid, Karim Mazaheri
- (2018)
Published **Modification of a plasma actuator model for induced wall-jet simulation**
Sharif Journal of Mechanical Engineering
Abbas Ebrahimi, Javad Omid
- (2018)
Published **Numerical Simulation of Plasma Actuator in Boundary Layer Separation Control by Utilizing the Modified Electrostatic Model**
Sharif Journal of Mechanical Engineering
Javad Omid, Karim Mazaheri
- (2017)
Published **Improving the performance of a numerical model to simulate the EHD interaction effect induced by dielectric barrier discharge**
International Journal of Heat and Fluid Flow
Javad Omid, Karim Mazaheri
- (2016)
Published **Simulation of DBD plasma actuator effect on aerodynamic performance improvement using a modified phenomenological model**
Computers & Fluids
Karim Mazaheri, Javad Omid, Kiarash Ch. Kiani

CONFERENCE PRESENTATIONS

- (2026)
Oral Presentation **Leveraging Angled Vibration to Induce Traveling Structured Phenomena In Vibrated Gas-Fluidized Granular Materials**
ASHG Annual Meeting, Orlando, Florida, US
Oscar Punch, Javad Omid, Rohit R Patil, Christopher Boyce
- (2025)
Oral Presentation **Structured Patterns in Granular Materials: From Bubble-Free to Bubbling Fluidization**
21st Annual Northeastern Granular Materials Workshop
Javad Omid, Christopher M. Boyce
- (2025)
Oral Presentation **Structured Bubbling Dynamics in Granular Media Induced by Oscillatory Gas Injection**
American Physical Society Annual Meeting, Anaheim, California, US
Javad Omid, Christopher M. Boyce

(2025) Oral Presentation	Faraday Waves in Granular Materials with Combined Vertical and Horizontal Vibrations American Physical Society Annual Meeting, Anaheim, California, US Javad Omid, Oscar J. Punch, Qiang Guo, Christopher M. Boyce
(2024) Oral Presentation	Structured Bubbling Flow in Granular Matter with Oscillated Gas Injection Which Alternates with Horizontal Position APS Division of Fluid Dynamics Meeting, Salt Lake City, Utah, US Javad Omid, Christopher M. Boyce
(2024) Oral Presentation	Oscillated Gas-Injection in Granular Matter: Exploring Structured Bubbling Flow AIChE Annual Meeting, San Diego, California, US Javad Omid, Christopher M. Boyce
(2024) Oral Presentation	Faraday Wave Patterns in Fluidized Granular Materials: Effects of Combined Vertical and Horizontal Vibration AIChE Annual Meeting, San Diego, California, US Javad Omid, Oscar J. Punch, Qiang Guo, Christopher M. Boyce
(2024) Oral Presentation	Descent of Intruders in Vibrated Gas Fluidized Beds AIChE Annual Meeting, San Diego, California, US Oscar J. Punch, Qiang Guo, Maty Gueye, Javad Omid, Christopher M. Boyce
(2023) Oral Presentation	Magnetic Resonance Imaging of Bubbles Injected into Liquid Suspensions AIChE Annual Meeting, Orlando, Florida, US Alireza Bordbar, Wasif Zia, Javad Omid, Janine Birnbaum, Einat Lev, Christopher M. Boyce
(2023) Oral Presentation	Magnetic resonance imaging of bubbles injected into liquid suspensions AIChE Annual Meeting, Las Vegas, Nevada, US Alireza Bordbar, Wasif Zia, Javad Omid, Janine Birnbaum, Einat Lev, Christopher M. Boyce
(2022) Oral Presentation	Periodically Structured Coalescence of Bubbles in a Vertical Stream Rising through Dense Suspensions APS Division of Fluid Dynamics Meeting, Indianapolis, Indiana, US Boyuan Chen, Azin Padash, Wasif Zia, Alireza Bordbar, Javad Omid, Christopher M. Boyce
(2022) Oral Presentation	DBD plasma actuator effect on mid-blade aerodynamics enhancement of a NREL 5-MW wind turbine International Conference on Futuristic Technologies in Control Systems & Renewable Energy (ICFCR) Javad Omid
(2020) Oral Presentation	Study of the Micro-Plasma Actuator Function in Improvement of Renewable Energy Technology 2nd Conference on Computational and Experimental Mechanics, Tehran, Iran Javad Omid
(2015) Oral Presentation	Numerical Simulation of Von-Karman Vortex Shedding Control by Utilizing Plasma Actuator 16th Fluid Dynamics Conference, Kermanshah, Iran Javad Omid, Karim Mazaheri
(2015) Oral Presentation	Electrostatic Simulation of DBD Plasma Actuator for Flow Separation Control around the Airfoil 23rd International Conference of Mechanical Engineering (ISME), Tehran, Iran Javad Omid, Karim Mazaheri, Kiarash Ch. Kiani

COMPUTER & PROGRAMMING SKILLS

Programming	C++ FORTRAN MATLAB	Python / R Programming (RStudio) Maple EES (Engineering Equation Solver)
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Software	COMSOL Multiphysics	3D Max
	ANSYS Fluent	SolidWorks
	ANSYS Meshing	Photoshop
	OpenFOAM	Microsoft Office
	GAMBIT	Tecplot
	AutoCAD	MFiX Multiphase Flow Simulation
	CATIA	Git/GitHub
	IBM SPSS Statistics	Notebook/JupyterLab
	ParaView	Linux/Unix

PROFESSIONAL SKILLS

Experimental Laboratory Skills	— Optical Imaging & High-Speed Camera Experiments
	— MRI-Based 3D Flow Imaging
	— Experimental Image Processing & Quantitative Analysis
	— Fluidized-Bed Design, Assembly & Operation
	— Gas–Solid & Liquid–Solid Multiphase Experiments
	— Granular Material Preparation & Characterization
	— Dense Suspension & Soft-Matter Preparation
	— Particle Tracking & Flow Visualization
	— Bubble Dynamics & Particle-Mixing Measurements
	— 3D Printing & Experimental Model Fabrication
Project Management & Leadership	— Experimental Setup Design & Instrumentation
	— Experimental Data Acquisition & Processing
	— Research Project Design
	— Grant & Proposal Writing
	— Academic Advising
Teaching, Mentorship & Academic Service	— Workshop & Training Program Organization
	— Thesis Supervision & Research Topic Definition
	— Teaching Assistance in Graduate & Undergraduate Courses
	— Curriculum & Content Development
	— Mentorship & Student Advising
Research & Analytical Skills	— Peer Review for International Known Journals
	— Conference Presentation & Workshop Facilitation
	— Computational Fluid Dynamics (CFD)
	— CFD–DEM & CFD–TFM Modeling
	— Multiphase Flow & Particle–Fluid Dynamics
	— Turbulent Flow Modeling
	— Granular Flow & Granular Materials
	— Fluidized-Bed Dynamics
	— Soft Matter & Complex Fluids
	— Multiphysics & Multiscale Modeling
	— Electrohydrodynamics (EHD) & Magnetohydrodynamics (MHD)
	— Aerodynamics & Boundary-Layer Flow
	— Flow Separation & Wake Control
	— Plasma & Micro-Actuator Flow Control
	— Renewable Energy & Wind Turbine Aerodynamics
	— Numerical Methods & Algorithm Development
	— Differential Evolution & Global Optimization
	— Parametric Optimization & Sensitivity Analysis
	— Model Calibration & Numerical Validation
	— Dimensionless Analysis & Engineering Correlations

Data Science & Machine Learning Skills	— Large-Scale & High-Dimensional Data Analysis
	— Data Processing, Integration & Visualization
	— Machine Learning & Predictive Modeling
	— Deep Learning & Neural Networks
	— Feature Selection & Feature Engineering
	— Classification & Regression
	— Autoencoders & Latent Representation Learning
	— Supervised & Unsupervised Learning
	— Multivariate Statistical Analysis
	— Statistical Modeling & Hypothesis Testing
	— Cross-Validation & Resampling
	— Regularized Regression (LASSO & Elastic Net)
	— Network & Graph-Based Data Analysis
	— Comparative & Heterogeneous Data Integration
	— Parameter Estimation & Optimization
	— Sensitivity & Uncertainty Analysis
	— ML-Driven Optimization
	— Automated Data-Processing Pipelines
	— Reproducible Python/R Workflows
	— High-Performance & Large-Scale Computing
Communication & Professional Skills	— Scientific Writing & Publication in High-Impact Journals
	— Effective Communication of Complex Data
	— Cross-functional Team Leadership
	— Interdisciplinary Teamwork & Leadership
	— Cross-Cultural & Multilingual Collaboration

PROFESSIONAL COURSES & CERTIFICATES

2025	Foundations of Project Management Google Career Certificate, US
2025	Principles of Management Saylor AcademySaylor Academy (Credential ID 8949223071JO)
2025	R Programming Johns Hopkins University, US (Credential ID Z4S9DX6LJ10E)
2024	Convolutional Neural Networks DeepLearning.AI, US (Credential ID XWQYAPUY3BE5)
2024	Neural Networks and Deep Learning DeepLearning.AI, US (Credential ID E0QCCZ6PUT4O)
2024	Advanced Learning Algorithms DeepLearning.AI, Stanford University, US (Credential ID FV3POQPHC286)
2024	Machine Learning Specialization DeepLearning.AI, Stanford University, US (Credential ID JHYA5KJI0ZYD)
2024	Python for Data Science and Machine Learning Bootcamp Udemy, US (Credential ID UC-3d001b17-6ac7-4f3e-9f88-747814acaa16)
2024	Supervised Machine Learning: Regression and Classification DeepLearning.AI, Stanford University, US (Credential ID 4OPDQSKOI9IR)
2024	Unsupervised Learning, Recommenders, Reinforcement Learning DeepLearning.AI, Stanford University, US (Credential ID K6N6BJDCVINA)
2024	Using Databases with Python University of Michigan, US (Credential ID F22M2AFDYNZH)
2024	Using Python to Access Web Data University of Michigan, US (Credential ID U8A4PPFP7FEC)

2024	Capstone: Retrieving, Processing, and Visualizing Data with Python University of Michigan, US (Credential ID WG5T4F6VLFQC)
2024	Python Data Structures University of Michigan, US (Credential ID SAJDAZ26PVSY)
2024	Data Analysis with Python Cognitive English Classes, India (Credential ID 40e38797349345398acc9e9561bfebbc)
2024	Python for Data Science Cognitive English Classes, India (Credential ID 198fc11bb6f44c3f9abeead6db6f9b4c)
2013	Advanced Computational Fluid Dynamics II (CFD II) Sharif University of Technology, Tehran, Iran
2012	Advanced Computational Fluid Dynamics I (CFD I) Sharif University of Technology, Tehran, Iran
2012	Advanced Mathematics Sharif University of Technology, Tehran, Iran

WORKSHOP PARTICIPATIONS

2025	21st Northeastern Granular Materials Workshop Yale University, CT, US
2023	18th Northeast Complex Fluids and Soft Matter Workshop Columbia University, NY, US
2022	17th Northeast Complex Fluids and Soft Matter Workshop Stevens Institute of Technology, NJ, US

PROFESSIONAL MEMBERSHIPS

SEP 2025 SEP 2024	American Institute of Chemical Engineers (AIChE) Professional Membership
AUG 2024 MAR 2022	American Physical Society (APS) Professional Membership

HONORS & AWARDS

FEB 2024	Journal Editor's Choice Issued by Clean Energy, Oxford University Press, UK Article titled "Advances and opportunities in wind energy harvesting using plasma actuators: a review" selected as Editor's Choice by Clean Energy, Oxford University Press
JAN 2022	Columbia University Full Scholarship Issued by Columbia University, New York City, NY, US Full scholarship for PhD study and research at School of Engineering
JAN 2021	DAAD Scholarship Issued by German Academic Exchange Service, Germany Full Scholarship (180k €) for Research at Technical University of Munich (TUM), Germany
SEP 2012	Top Rank in Nationwide Entrance Examination for MS Studies Issued by National Organization Educational Testing, Iran Top 1% in Nationwide Entrance Examination for graduate studies (MS) among more than 3,100 participants

SEP 2007	Top Rank in Nationwide Entrance Examination for BS Studies Issued by National Organization Educational Testing, Iran Top 1% in Nationwide Entrance Examination for undergraduate studies (BS), among more than 270,000 participants
SEP 2006	4th-Rank Student in Diploma Issued by Markazi Province Educational Office, Arak, Iran 4th-rank student in high school diploma at the city of Arak, Iran, among 1,177 students

LANGUAGE PROFICIENCY

Native	Persian Language
Full Professional Proficiency	English Language
Limited Working Proficiency (A2)	French Language
Limited Working Proficiency (A2)	German Language
Limited Working Proficiency	Arabic Language

REFERENCES

Christopher Boyce

Associate Professor
Columbia University, New York City, NY, US
Email: cmb2302@columbia.edu
Website: boyce.cheme.columbia.edu

Karim Mazaheri

Professor
Sharif University of Technology, Tehran, Iran
Email: mazaheri@sharif.edu
Website: ae.sharif.edu/~portal/faculty/2074029919